

1. Compute the total time required to transfer a file of 2.5 MB assuming RTT of 55ms and the packet size is 2KB and an initial delay of one RTT of handshaking before the data is sent on the line. If the link bandwidth is 20Mbps, and after sending every data packet we have wait one RTT before sending the next, compute the total time taken to transfer the complete file.

Ans. wait one RTT before sending the next:

Initial Handshaking = $1 * R.T.T. = 1 * 55 \text{ ms} = 55 \text{ ms}$

Inter packet gap = $1 R.T.T. = 55 \text{ ms}$

Total number of packets = $2.5\text{MB}/2\text{KB} = 1250 \text{ PACKETS}$

time to send 1 packet is $2\text{KB} / 20 \text{ Mbps} = 0.8 \text{ ms}$

Total Time = T Initial Handshaking time + 1250 packets T.T. + 1249 * R.T.T. (waiting time)

$$= 55 + 1250 * 0.8 + 1249 * 55 = 55 + 1250 + 68695 = 69750 \text{ ms} = 69.75 \text{ sec}$$

2. Calculate the bandwidth*delay product for the following links. Use one-way delay, measured from first bit sent to first bit received.

a. 100Mbps Ethernet with a delay of 10μsec.

b. 10Mbps Ethernet with a single store-and-forward switch in the path, and packet size of 50000 bits. Assume that each link introduces a propagation delay of 10 μsec and each switch is retransmitting immediately after it has finished receiving the packet.

c. 2 Mbps T1 link, with a transcontinental on-way delay of 50ms.

Ans. a. The b x w product is $100 \times 10^6 \text{ bits/sec} \times 10 \times 10^{-6} \text{ secs} = 1000 \text{ bits} = 125 \text{ bytes}$

b. The first-bit delay is 5020 microsecs through the store-and-forward switch. The bandwidth x delay product is thus $10 \text{ Mbps} \times 5020 \text{ microsecs} = 50200 \text{ bits}$ (Alternatively, you can think of it as each link can hold 100 bits and the switch can hold 50000 bits.)

c. $2 \times 10^6 \text{ bits/sec} \times 50 \times 10^{-3} \text{ sec} = 1,00,000 \text{ bits} = 12500 \text{ bytes}$

3. What is the total delay (Latency) for a frame of size 5million bits that is being on the sent on a link with 10 routers each having a queuing time of 2 μsec and a processing time of 1μsec. The link has a bandwidth of 5Mbps. Which component of the total delay is dominant? Which one is negligible? Assume The length of the link is 2000 Km. The speed of light inside the link is $2 \times 10^8 \text{ m/s}$

Ans. Propagation time = distance / propagation speed = $2000 \text{ Km} / 2 \times 10^8 \text{ m/s} = 10 \text{ ms}$

Transmission time = Message size / Bandwidth = 5×10^6 bits / 5 Mbps = 1 s

Queuing time = 10 routers * 2 us = 20 us

Processing Delay = 10 routers * 1 us = 10 us

Total delay (latency) = 10 ms + 1 s + 20 us + 10 us = 1010.03 ms = 1.01003 s / 1 s

4. Compute two-dimensional parity for the given data word 1100 1111 0110 0011 1110 0001 0011 01101000 (four bits at a time) using even parity. Introduce error in two different bit positions and correct the errors.

Ans.	1100	0
	1111	0
	0110	0
	0011	0
	1110	1
	0001	1
	0011	0
	0110	0
	1000	1
	0100	1

Introduce error in two different bit positions and repeat the same process.

Error Case 1: Flip bit at Row 5, Column 3

Original R5: 1110

Erroneous R5: 1100

Detection

Row 5 parity fails

Column 3 parity fails

☒ Error located at (R5, C3)

✓ Flip bit back → corrected

Error Case 2: Flip bit at Row 9, Column 1

Original R9: 1000

Erroneous R9: 0000

Detection

Row 9 parity fails

Column 1 parity fails

☒ Error located at (R9, C1)

✓ Flip bit back → corrected

Final Corrected Data (Recovered)

1100 1111 0110 0011 1110 0001 0011 0110 1000

Note: two-dimensional parity catches all 2-bit errors. If the 2 bits that have been altered are not in the same column, then the column parity bits will be incorrect, and if they are not in the same row, the row parity bits will be incorrect. Two positions cannot simultaneously be in the same row and same column; hence the error can be detected.

5. A network with pure ALOHA transmits 250 bits per frame on a shared channel of 500 kbps. Compute the time required to make the transmission collision free?

Ans. Average Frame Transmission time T_{fr} is 250 bits/500 kbps or 0.5 ms.

The vulnerable time is $2 \times 0.5\text{ms} = 1\text{ms}$.

This means no station should send later than 0.5 ms before this station starts transmission and no station should start sending during the one 0.5- ms period that this station is sending.

6. Compute total time delay for transmitting a file of 5MB file on link in 50Mbps line of length of 12000km where the signal travel with a velocity of 2.4×10^8 m/sec.

Ans. Propagation time= $12000 \times 1000 / 2.4 \times 10^8 = 50\text{ms}$

Transmission time= $5\text{MX}8 \text{ bits} / 50\text{Mbs} = 0.8\text{sec}$

Total time= Propagation time+ Transmission time= $50\text{ms} + 800\text{ms} = 850\text{ms}$

7. Consider the message sender wants to send is 1010001101, and the generator polynomial is $x^5 + x^4 + x^2 + 1$. Find the message transmitted by the sender. If the receiver receives the message, check if the receiver receives the correct message or not.

Ans. **Given Message (M)** = 1010001101, **Generator polynomial**

$$G(x) = x^5 + x^4 + x^2 + 1$$

Generator in binary (from x^5 to x^0):

$$G = 110101 (\text{degree} = 5)$$

Step 1: Append 5 zeros to the message

101000110100000

Step 2: Perform modulo-2 division (XOR) with generator 110101

After binary division, the **remainder (CRC)** obtained is: 01110

Step 3: Transmitted message

Transmitted frame = Message + CRC: 1010001101 01110

Step 4: Receiver-side checking

The receiver divides the received message 101000110101110 by the generator 110101.

The **remainder** = 00000

- **CRC bits** = 01110
- **Message transmitted by sender** = 101000110101110
- **Receiver check result:** Remainder is zero → **No error detected**

Correct message received

- If remainder $\neq 0$ → error detected
- If remainder = 0 → frame assumed error-free

8. An image of 1024*768 pixels with 3bytes/pixel. Assume the image is uncompressed. How long does it take to transmit it over a 56kbps modem channel? Over a 1-Mbps cable modem? Over a 10Mbps ethernet? over 100Mbps ethernet

Ans. The image is 1024*768*3 bytes or 2,359,296 bytes. This is 18,874,368 bits.

At 56,000 bits/sec, it takes about 337.042 sec.

At 1,000,000 bits/sec, it takes about 18.874 sec.

At 10,000,000 bits/sec, it takes about 1.887 sec.

At 100,000,000 bits/sec, it takes about 0.01887 sec.

9. Consider a source computer S transmitting a file of size 10^6 bits to a destination computer D over a network of two routers R1 and R2 and three links L1, L2 and L3. L1 connects between S and R1; L2 connects R1 and R2, and L3 connects R2 and D. Let length of each link be 100km and signal travels at a speed of 10^8 m/sec. If the file broken into 1000 packets each of size 1000bits. Find the total sum of transmission and propagation delays in transmitting the file from S to D. Assume the bandwidth of the channel is 1Mbps.

Ans.

Data:

Number of packets = $n = 1000$

Size of each Packet = $L = 1000$ bits

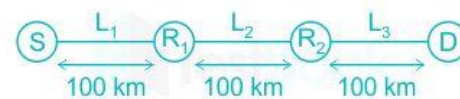
The link bandwidth on each link is = $BW = 1\text{Mbps} = 10^6$ bits/s

Each link be of length = $d = 100$ km

Signals travel over each link at a speed = $v = 10^8$ m/s

Transmission delay = T

Propagation delay = P



Formula:

$$T = \frac{L}{BW}$$

$$P = \frac{d}{v}$$

Propagation delay to travel from S to R1 (P) = (Distance) / (Link Speed) = $10^5 / 10^8$
= 1ms

Total propagation delay to travel from S to D = $3 \times 1 \text{ ms} = 3\text{ms}$

Total Transmission delay for 3 packets (T) = $3 \times (\text{Number of Bits}) / \text{Bandwidth} = 3 \times (1000 / 10^6) = 3\text{ms}$.

So, the total time taken to transmit the first packet from source to destination is $P+T = 3+3 \text{ ms} = 6\text{ms}$

The first packet will take 6ms to reach D.

While first packet was reaching D, other packets must have been processing in parallel. So, D will receive remaining packets 1 packet per 1 ms from R2. So, remaining 999 packets will take 999 ms. And total time will be $999 + 6 = 1005 \text{ ms}$

10. Calculate the latency (from first bit sent to last bit received) for the following: (a) 10-Mbps Ethernet with a single store-and-forward switch in the path, and a packet size of 5000 bits. Assume that each link introduces a propagation delay of $10 \mu\text{s}$

and that the switch begins retransmitting immediately after it has finished receiving the packet. (b) Same as (a) but with three switches.

Ans. a) transmission time = $5000/10 \times 10^6 = 500 \text{ us}$

packet reaches at switch = $500 + 10 = 510 \text{ us}$

then from switch to destination = 510 us

so total latency = 1020 us

b) for 3 switches = $510 \times 4 = 2040 \text{ us}$ (as packet must go over 4 links)

11. Suppose the following block of 32 bits is to be sent using checksum 8 bits 11100101011010111000111101011011 and check your answer for the receiver receives 11100101111010111000111101011011.

Ans.	Bits
W1	11100101
W2	01101011
W3	10001111
W4	01011011

Step 2: 1's complement addition

Add W1 + W2

```
11100101
+01101011
-----
```

1 01010000 (carry generated)

Add carry: $01010000 + 1 = 01010001$

Add W3

```
01010001
+10001111
-----
```

1 11100000

Add carry: $11100000 + 1 = 11100001$

Add W4

```
11100001
+01011011
-----
```

1 00111100

Add carry: $00111100 + 1 = 00111101$

Step 3: Take 1's complement → Checksum

$00111101 \rightarrow 11000010$

Sender transmits 11100101 01101011 10001111 01011011 11000010

Receiver Side: Error Checking

Received block: 11100101 11101011 10001111 01011011 11000010

(Notice **second byte changed** → error introduced)

Step 1: Add all received 8-bit words

11100101

11101011

10001111

01011011

11000010

Performing 1's complement addition → **final sum ≠ 11111111**

- Final sum is **not all 1s**
- Therefore, **checksum test fails**
- **Checksum computed at sender:** 11000010
- **Receiver checksum result:** Error detected
- **Conclusion:** The received message is **incorrect**

General definitions

Latency (Delay)

- The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.
- Latency = propagation time + transmission time + queuing time + processing delay
- **Propagation Time**
 - The time required for a bit to travel from the source to the destination.
$$\text{Propagation time} = \text{Distance} / \text{Propagation speed}$$
- The propagation speed of electromagnetic signals depends on the medium and on the frequency of the signal. For example, in a vacuum, light is propagated with a speed of 3×10^8 m/s. It is lower in air; it is much lower in cable.